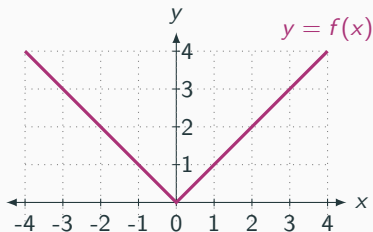


Warm up: Limit Definition of the Derivative

Definition

If the limit exists, **the derivative** $f'(a)$ of a function $y = f(x)$ at a point $x = a$ is the limit:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$



Discussion Question

Using the limit definition, discuss how to find the following derivatives, if they exist:

(a) $f'(2) =$

(b) $f'(-1) =$

(c) $f'(0) =$

Lecture 12

Derivatives and Limits: Focus on Theory Shortcuts to Differentiation

Big Questions

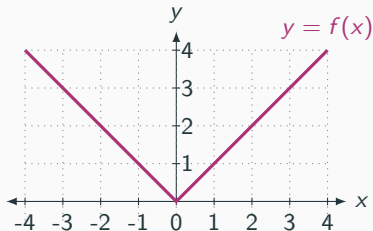
- What does it mean for a function to be continuous?
- If we have a formula for a function, can we find the formula for its derivative?

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Continuity

Definition

The function f is **continuous** at $x = c$ if f is defined at $x = c$, $\lim_{x \rightarrow c} f(x)$ exists, and

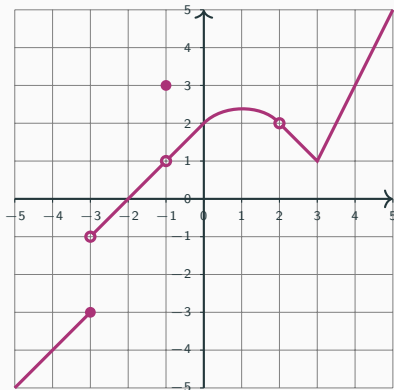
$$\lim_{x \rightarrow c} f(x) = f(c).$$

A function is continuous on an interval (a, b) if it is continuous at every point in the interval.

Poll Question

The graph of $y = f(x)$ is to the right. At which values of x below is the function f continuous?

- A** -4 **B** -3 **C** -1 **D** 2

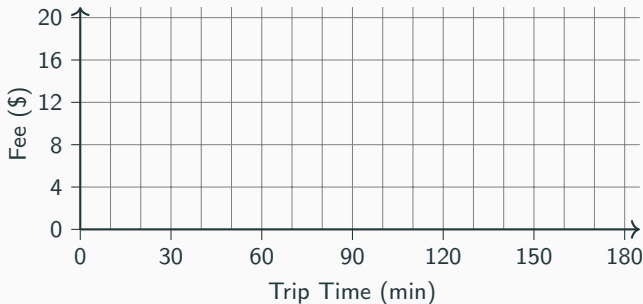


Bike Share Toronto Fees

If you have an annual membership with Bike Share Toronto, bike trips of up to 30 minutes are free. However, if your trip lasts for longer than 30 minutes and up to an hour, a fee of \$4 will be charged. If your trip lasts for longer than 60 minutes and up to 90 minutes, you'll be charged \$8. This pattern continues for each 30 minutes, up to 3 hours.

Discussion Question

Graph the fees as a function f of trip time on the grid below. Mia returned her bike after a 30min 15sec ride and thought the fee was unfair. How can you use continuity to explain why Mia thought the pricing was unfair?



If we have a formula for a function, can we find the formula for its derivative?

We use the word “differentiation” to describe the process of finding a formula for the derivative of a function.

As we approach this topic,

- focus on understanding the logic behind the formulas;
- focus on the relevance in applications,
- use the “rule of 4” by thinking in each of the four representations:

Verbal	Graphs
Data	Formulas

Comparing Stocks

The closing values (\$) of three stocks f , g , and h , over a 5-day period are tabulated below.

Verbal	Graphs
Data	Formulas

t (days)	1	2	3	4	5
$f(t)$ (\$)	6	6	6	6	6
$g(t)$ (\$)	80	80	80	80	80
$h(t)$ (\$)	300	300	300	300	300

Poll Question

According to the data above, how do the derivatives f' , g' , h' likely relate to each other?

A $f'(t) > g'(t) > h'(t)$

B $f'(t) < g'(t) < h'(t)$

C $f'(t) = g'(t) = h'(t)$

Derivative of a Constant Function

Let c be any number.

If $f(x) = c$ is a constant function, then $f'(x) =$

In Leibniz notation, if $y = f(x)$, then

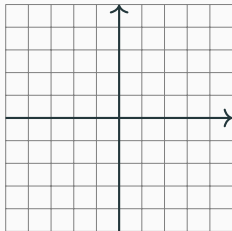
Derivative of a Linear Function

Discussion Question

For each of the functions below, sketch the graph, then use the graph to find the derivative.

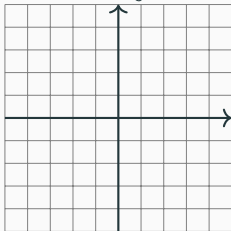
Verbal	Graphs
Data	Formulas

$$f(x) = 2x - 3$$



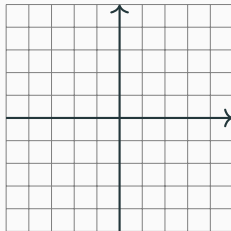
$$f'(x) =$$

$$f(x) = -\frac{1}{3}x + 2$$



$$f'(x) =$$

$$f(x) = mx + b$$



$$f'(x) =$$

Conclusion: For linear functions, the derivative is

Which functions can we differentiate?

So far, we've seen how to differentiate:

- ☐ constant functions
- ☐ linear functions

What other types of functions might we want to learn how to differentiate?

If $f(x) = x^n$, can we find a formula for $f'(x)$?

Verbal	Graphs
Data	Formulas

Poll Question

Let's consider the case $n = 2$. Let $y = f(x) = x^2$.

- (a) According to the limit definition of derivative, which of the following expressions is equal to $f'(x)$?

A $\lim_{h \rightarrow 0} \frac{(x^2 + h^2) - x^2}{h}$

B $\lim_{h \rightarrow 0} \frac{(x + h)^2 - h^2}{h}$

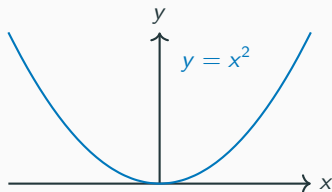
C $\lim_{h \rightarrow 0} \frac{(x + h)^2 - x^2}{h}$

- (b) **Use the limit definition** of the derivative to find a formula for $f'(x)$.

Recall: The limit definition of the derivative says that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

(More space for (b).)



It is possible to use a similar line of reasoning for $f(x) = x^n$ for any whole number n to find

$$f'(x) = \boxed{}$$

In fact, this result is true for any **real** number n .

Power Rule: For any real number n ,

$$\frac{d}{dx}(x^n) = \boxed{}$$

Discussion Question

Using only the Power Rule, find the following derivatives.

(a) $\frac{d}{dx}(x^8)$

(b) $\frac{d}{dx}\left(\frac{1}{x^2}\right)$

(c) $\frac{d}{dx}(\sqrt{x})$

Summary and Reflection

Write down a few functions that we **can** quickly differentiate, based on today's class.

Write down a few functions that we **cannot yet** quickly differentiate, based on today's class.

To do before next class:

- Complete question 1 from Week 4 Checkpoint Workbook.
- Finish your reflection and any remaining reading and homework from Week 3 by 10pm tomorrow.
- Start Homework 4 (questions 1-8).
- The first draft of Project 1 is due at 10pm tomorrow night.

Have a great day!